1.02 Complex Numbers: Quadratic Equations

Worked Solutions

1 a)

On line 2: He doesn't halve the -2 inside the bracket when completing the square.

On line 3: He subtracted 4 from the RHS rather than adding it and doesn't put a $\pm$ in front of the $\sqrt{-9}$

On line 4: He didn't square root the 9

1 b)

$$\begin{aligned} z^2 - 2z + 5 &= 0 \\ (z-1)^2 - 1 &= -5 \\ z - 1 &= \pm\sqrt{-4} \\ z &= 1 \pm 2i \end{aligned}$$

2 a)

$$\begin{aligned} z^2 &= -1 \\ z &= \pm\sqrt{-1} \\ z &= \pm i \end{aligned}$$

2 b)

$$\begin{aligned} z^2 &= -10 \\ z &= \pm\sqrt{-10} \\ z &= \pm\sqrt{10}i \end{aligned}$$

2 c)

$$\begin{aligned} (z-1)^2 - 1 &= -17 \\ (z-1)^2 &= -16 \\ z - 1 &= \pm\sqrt{-16} \\ z &= 1 \pm 4i \end{aligned}$$

2 d)

$$\begin{aligned}
4\left(z-\frac{1}{2}\right)^2-1&=-17\\
4\left(z-\frac{1}{2}\right)^2&=-16\\
\left(z-\frac{1}{2}\right)^2&=-4\\
z-\frac{1}{2}&=\pm\sqrt{-4}\\
z&=\frac{1}{2}\pm 2i
\end{aligned}$$

2 e)

$$\begin{aligned}
2\left(z-\frac{1}{2}\right)^2-\frac{1}{2}&=-1\\
2\left(z-\frac{1}{2}\right)^2&=-\frac{1}{2}\\
\left(z-\frac{1}{2}\right)^2&=-\frac{1}{4}\\
z-\frac{1}{2}&=\pm\sqrt{-\frac{1}{4}}\\
z&=\frac{1}{2}\pm\frac{1}{2}i
\end{aligned}$$

2 f)

$$\begin{aligned}
\left(z-\frac{1}{2}\right)^2-\frac{1}{4}&=-3\\
\left(z-\frac{1}{2}\right)^2&=-\frac{11}{4}\\
z-\frac{1}{2}&=\pm\sqrt{-\frac{11}{4}}\\
z&=\frac{1}{2}\pm\frac{1}{2}\sqrt{11}i
\end{aligned}$$

3 a)

$$\begin{aligned}
(z+5)^2&=-7\\
z+5&=\pm\sqrt{-7}\\
z&=-5\pm\sqrt{7}i
\end{aligned}$$

3 b)

$$\begin{aligned}
(z-1)^2&=-12\\
z-1&=\pm\sqrt{-12}\\
z&=1\pm\sqrt{12}i\\
z&=1\pm 2\sqrt{3}i
\end{aligned}$$

3 c)

$$\begin{aligned} 2(z+4)^2+8 &= 6 \\ 2(z+4)^2 &= -2 \\ (z+4)^2 &= -1 \\ z+4 &= \pm\sqrt{-1} \\ z &= -4 \pm i \end{aligned}$$

3 d)

$$\begin{aligned} 8(z+5)^2-3 &= -9 \\ 8(z+5)^2 &= -6 \\ (z+5)^2 &= -\frac{3}{4} \\ z+5 &= \pm\sqrt{-\frac{3}{4}} \\ z &= -5 \pm \frac{\sqrt{3}}{2}i \end{aligned}$$

4)

$$\begin{aligned} z^4+2z^2-3 &= 0 \\ (z^2+1)^2-1 &= 3 \\ (z^2+1)^2 &= 4 \\ z^2+1 &= \pm 2 \\ z^2 &= 1 \text{ or } z^2 = -3 \\ z &= \pm 1 \text{ or } z = \sqrt{-3} = \pm\sqrt{3}i \end{aligned}$$

5 a)

$$\begin{aligned} & z^2-(1+i+1-i)z+(1+i)(1-i) \\ &= z^2-z+iz-z-iz+1+i-i+1 \\ &= z^2-2z \end{aligned}$$

5 b)

$$\begin{aligned} & z^2-(3-i+3+i)z+(3+i)(3-i) \\ &= z^2-3z+3iz-3z-3iz+9+3i-3i+1 = z^2-6z+10 \end{aligned}$$

5 c)

$$\begin{aligned} & z^2-(1+2i+1-2i)z+(1+2i)(1-2i) \\ &= z^2-z+2iz-z-2iz+1+2i-2i+4 = z^2-2z+5 \end{aligned}$$

5 d)

$$z^2 - (-4 + 3i - 4 - 3i)z + (-4 + 3i)(-4 - 3i)$$
$$= z^2 + 4z - 3iz + 4z + 3iz + 16 + 12i - 12i + 9 = z^2 + 8z + 25$$

5 e)

$$z^2 - (i - i)z + (i)(-i)$$
$$= z^2 - iz + iz + 1 = z^2 + 1$$

5 f)

$$z^2 - (5i - 5i)z + (5i)(-5i)$$
$$= z^2 - 5iz + 5iz + 25 = z^2 + 25$$

5 g)

$$z^2 - (-1 + \sqrt{2}i - 1 - \sqrt{2}i)z + (-1 + \sqrt{2}i)(-1 - \sqrt{2}i)$$
$$= z^2 + z - \sqrt{2}iz + z + \sqrt{2}iz + 1 - \sqrt{2}i + \sqrt{2}i + 2 = z^2 + 2z + 3$$

6 a)

$$3 + 8i$$

6 b)

$$(z - (3 + 8i))(z - (3 - 8i)) = z^2 - (3 + 8i)z - (3 - 8i)z + (3 + 8i)(3 - 8i)$$
$$= z^2 - 3z - 8iz - 3z + 8iz + 9 + 24i - 24i + 64 = z^2 - 6z + 73$$

So $p = -6$ and $q = 73$

7 a)

$$-\frac{1}{2} + 4i$$

7 b)

$$\left(z - \left(-\frac{1}{2} - 4i\right)\right)\left(z - \left(-\frac{1}{2} + 4i\right)\right)$$
$$= z^2 - \left(-\frac{1}{2} - 4i\right)z - \left(-\frac{1}{2} + 4i\right)z + \left(-\frac{1}{2} - 4i\right)\left(-\frac{1}{2} + 4i\right)$$
$$= z^2 + \frac{1}{2}z + 4iz + \frac{1}{2}z - 4iz + \frac{1}{4} - 2i + 2i + 16$$
$$= z^2 + z + \frac{65}{4}$$

So $p = 1$ and $q = \frac{65}{4}$

8 a)

$$a - bi$$

8 b)

$$\begin{aligned}(z-(a+bi))(z-(a-bi)) &= z^2-(a+bi)z-(a-bi)z+(a+bi)(a-bi)\\ &= z^2-az-biz-az+biz+a^2+abi-abi+b^2 = z^2-2az+a^2+b^2\end{aligned}$$

So $p = -2a$ and $q = a^2 + b^2$

9 a)

$$\begin{aligned}z^2-4z+7&=0\\(z-2)^2-4&=-7\\z-2&=\pm\sqrt{-3}\\z&=2\pm\sqrt{3}i\end{aligned}$$

9 b)

$$|2+\sqrt{3}i| = \sqrt{4+3} = \sqrt{7}$$

So $p = \sqrt{7}$

9 c)

Area is 7π units^2

9 d)

$$w = -2+\sqrt{3}i \text{ or } w = -2-\sqrt{3}i$$

9 e)

$$\begin{aligned}(w-(-2+\sqrt{3}i))(w-(-2-\sqrt{3}i)) &= w^2+4w+(-2+\sqrt{3}i)(-2-\sqrt{3}i)\\ &= w^2+4w+7\end{aligned}$$

$a = 4$, $b = 7$

10)

Distinct complex roots: $b^2 - 4ac < 0$

$$\begin{aligned}k^2-4\times4&<0\\(k-4)(k+4)&<0\\\therefore -4<k&<4\end{aligned}$$

11)

$$\begin{aligned} az^2 + c &= 0 \\ z^2 &= -\frac{c}{a} \\ z &= \pm\sqrt{-\frac{c}{a}} \end{aligned}$$

$\therefore$ it only has purely imaginary roots

12)

$$\begin{aligned} az^2 + bz &= 0 \\ z(az + b) &= 0 \end{aligned}$$

The roots are $z = 0$ and $z = -\frac{b}{a}$ which are real